



Project Title: CareMate

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Year grade: 5

School: Northside Montessori School, Pymble

State event: NSW

Please answer to the best of your abilities

1. Project summary (*max 200 words*)

CareMate is an AI-powered home-assistant robot designed to support elderly people living alone by improving safety, emotional wellbeing, and cognitive engagement. These are common concerns related to elderly wellbeing. The robot is built using Raspberry Pi 4B running Python on Linux, a Raspberry Pi Pico 2WH microcontroller running MicroPython, multiple sensors, electronic components and AWS (Amazon web services) AI functionality.

CareMate provides nighttime safety support by detecting movement and turning on soft LED path lighting to guide the person without bright lights. It can also help remove small obstacles using its gripper. A Hall sensor detects if a door is left open, while a heat sensor checks for dangerous appliance heat; CareMate then alerts the user using natural speech generated with AWS Polly.

For emotional support, the robot uses camera images and AWS Rekognition to detect sad facial emotion, then wirelessly triggers family photos, videos, or music on the TV. CareMate also runs a cognitive game using images, quizzes, speech input, and Amazon Lex V2 to understand responses and talk back naturally.

2. TECHNOLOGIES USED

1. Raspberry Pi 4B running Python3 on Linux

The main computer running on the robot powered by two 18650 batteries.

2. Raspberry Pi Pico 2WH running MicroPython

Microcontroller that controls electronic components like the LED strip by receiving serial commands from laptop and sensors like heat sensor and hall sensor.

3. Freenove Robot Kit for Raspberry Pi

The kit which I used to construct the robot.

1. Freenove HAT

The interface between the Raspberry Pi and the robot.

2. Servo x 2 + gripper arm

A two-servo arm capable of picking up objects.

3. 18650 battery holder for the batteries that powers everything.

4. DC motors X 2 to move the robot.

4. Artificial Intelligence (AI) - Raspberry Pi Official Camera and Amazon Web Services)

Rekognition API

Mounted on the robot, camera stream that sends image/video content which is used to call AWS (Amazon Web Services) Rekognition API for emotion detection on face.

5. Windows Surface Book 2

The second computer running python3, acts as a wireless receiver of commands from Raspberry Pi. It is used to play songs and video on TV connected through HDMI. It also sends control signals to the Raspberry Pi Pico and receives data from it through a serial port. This is to control electronic components like LED strip and get sensor data attached to Raspberry Pi Pico.

6. 18650 batteries x2

Powers the robot and the Raspberry Pi.

7. Artificial Intelligence (AI) - Jabra 710 Speaker and Amazon Web Services Polly API

A speaker and microphone that uses wireless Bluetooth connection to Raspberry Pi. For the text-to-speech it calls AWS (Amazon Web Services) AWS Polly API to speak in natural language. Microphone is used for interpreting natural language and extracting intent.

8. Ultrasonic distance sensor

Can measure the distance up to 300cm.

9. Mobile phone connected to Raspberry Pi over wifi. It is used as small screen to build face of robot.

3. Additional information (*optional*)

*Insert images, screenshots or any links to your project if published online **or if you have video demonstration** in this section in support of your submission.*

How did you choose your project and What inspired the idea for your project?

My grandmother in India lives alone and has arthritis. I got this idea by looking at her daily challenges and I wanted to use technology to make a difference.

What challenges or problems did you encounter?

- a. Wireless connection
- b. AWS access and policy for APIs
- c. Building the body of robot to support everything.

Link to photos and video -

<https://github.com/superuser5698/CareMate/tree/main>

Describe the design and key features of your project

Features:

1. Nighttime help for elderly safety
 - a. If during the night if the person wakes up to go to the toilet, then the robot will detect it using its distance/motion and then it turns on a strip of LED lights to guide the person towards toilet without disturbing the sleep with bright light. It can remove objects lying on the floor to avoid the risk of tripping over which can be serious for elderly person.
2. Door left open and dangerous heat level detection: avoid the safety hazard if elderly person forgets things by mistake.
 - a. For the door detection I used a Hall sensor with a magnet on the door and if door is left accidentally open in the night for more than 5 minutes then it sends signal to Pico microcontroller which then transmits data to Raspberry Pi (Robot) who then alerts the person in a nice, friendly voice using AWS text to speech API.
 - b. If some appliance like cooktop is accidentally left on, then temperature sensor will detect excessive heat sends signal to Pico microcontroller which then transmits data to Raspberry Pi (Robot) who then alerts the person in a nice, friendly voice using AWS text to speech API.

3. Emotional support for elderly - Emotion recognition – Artificial intelligence
 - a. Robot uses camera and AWS Amazon Rekognition face emotion detection API to check if the person is looking sad. If person is sad, then raspberry pi sends commands to Windows laptop over wifi using socket to a song, video or photos of their grandchildren or relatives to cheer them up.
4. Brain exercise to help with Cognitive decline – Artificial intelligence
 - a. To reduce the decline of persons cognitive ability and help them remember things it, robot plays an interesting and fun game with elderly person. It shows some images, quiz etc on mobile phone screen (Robot's face) and asks user to identify a person or answer the quiz. It listens to user response and passes on the audio to Amazon Lex V2 API to extract intent/text and talks back to the person accordingly.